

# Model Optimization and Tuning Phase Template

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## Name

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## Project Title

Harnessing EC2, VPC & MobaXterm for Cutting-Edge Entity Hosting

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## Model Optimization and Tuning Phase

The Model Optimization and Tuning Phase in this project focuses on improving the performance, security, and efficiency of the cloud infrastructure deployed using AWS services. Instead of machine learning models, this phase involves optimizing system configurations such as instance selection, network setup, and remote connectivity to achieve better performance and reliability.

The project was designed to deploy and manage a cloud-based hosting environment using Amazon EC2 and VPC, while ensuring secure and efficient remote access through MobaXterm.

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## System Optimization Parameters

### Component 1: EC2 Instance Optimization

- **Instance Type Selection:** Different instance types such as t2.micro and t2.small were considered based on performance and cost.
  - **Storage Configuration:** Optimized storage allocation to balance cost and performance.
  - **Security Groups:** Configured to allow only required ports (e.g., SSH - port 22, HTTP - port 80).
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### Component 2: VPC Configuration

- **Subnet Design:** Public subnet used for hosting EC2 instance.
  - **Internet Gateway:** Configured to enable internet access.
  - **Routing Tables:** Proper routing rules applied for secure communication.
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## Hyperparameter Tuning Documentation

### Component 3: Remote Access Optimization (MobaXterm)

- **SSH Configuration:** Tested secure key-based authentication instead of password login.
  - **Connection Stability:** Optimized session settings for faster and stable connectivity.
  - **File Transfer:** Efficient transfer of files between local system and EC2 instance.
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## Component 4: Performance Optimization

- **Server Setup:** Installed and configured web server (Apache/Nginx).
  - **Command Optimization:** Used efficient Linux commands for faster setup and deployment.
  - **Resource Monitoring:** Checked CPU and memory usage to avoid overloading.
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## Final System Selection Justification

### Final Setup

EC2 + VPC + MobaXterm Configuration

### Reasoning

The selected cloud setup provided a balance between performance, security, and cost-effectiveness. EC2 enabled scalable computing, VPC ensured secure network isolation, and MobaXterm allowed efficient remote management. This configuration was simple to implement, reliable, and suitable for real-world cloud hosting scenarios.